

1. Administrative & Study Identification

Full Study Title: Real-world Outcomes Among Patients With Melanoma Treated With Neoadjuvant Nivolumab+Relatlimab or Nivolumab+Ipilimumab
Short Title/Acronym: Real-world Neoadjuvant Nivo+Rela / Nivo+Ipi in Melanoma
Protocol Number: CA224-1084
NCT Number: NCT07091695
Posting ID: 70087010672967855 **Sponsor/Company Name:** Bristol Myers Squibb
Investigational Product: Nivolumab (drug name) + Relatlimab (Opdualag) or Nivolumab + Ipilimumab (Opdivo + Yervoy)
Phase of Development: Not Applicable (Observational / Real-World Study)
Trial Status: COMPLETED **Medical Monitor:** Bristol Myers Squibb Study Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe the baseline demographics, clinical characteristics, and treatment history among patients with melanoma treated with neoadjuvant nivolumab+relatlimab or nivolumab+ipilimumab. **Secondary Objectives:** To evaluate real-world clinical outcomes including Overall Survival (OS), Time to Treatment Discontinuation (TTD), Duration of Treatment (DOT), Pathologic Complete Response (pCR), Event-Free Survival (EFS), Distant Metastasis-Free Survival (DMFS), and adverse events (AEs) over a follow-up of up to 42 months.

2.2 Background & Rationale To address the knowledge gap regarding the real-world effectiveness and treatment patterns in adult patients diagnosed with clinically palpable stage III resectable melanoma in the U.S. managed by physicians in Cardinal Health's OPEN network. This study generates real-world evidence to better understand the impact, sequence timing, and dose modifications of neoadjuvant dual-immunotherapy combinations outside the controlled environment of clinical trials.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective Cohort Study) **Control Type:** N/A (Cohort assessment) **Blinding:** Open-label / Unblinded (Retrospective chart review)
Randomization: N/A

3.2 Planned Enrollment & Duration Target \$\$\$: 100 adult patients
Number of Sites: U.S. sites (managed by physicians in Cardinal Health's OPEN network) **Estimated Study Start:** 11-May-2025
Estimated Primary Completion: 06-Nov-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants at least 18 years old at diagnosis of clinically palpable stage III resectable melanoma. 2. Participants treated with nivolumab+relatlimab (nivo+rela) or nivolumab+ipilimumab (nivo+ipi) in the neoadjuvant setting. 3. Treated with nivolumab+relatlimab (nivo+rela) or nivolumab+ipilimumab (nivo+ipi) in the neoadjuvant setting from 18 March 2022 onwards. 4. Participants with at least 6 months of follow-up from initiation of neoadjuvant therapy, unless deceased prior to 6 months of follow-up.

4.2 Exclusion Criteria 1. Participants received nivo+rela or nivo+ipi for clinically palpable stage III resectable melanoma as part of a therapeutic clinical trial. 2. Participants had history of diagnosis of any malignancy (except for non-melanoma skin cancer) in the last 2 years. 3. Participants received adjuvant nivo+rela or adjuvant nivo+ipi after neoadjuvant nivo+ipi. 4. Participants received adjuvant nivo+ipi after neoadjuvant nivo+rela. 5. Participants received any systemic therapy prior to the initiation of neoadjuvant nivo+rela or nivo+ipi.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product: Nivolumab
ATC Code: L01FF01 **Dosage/Frequency:** Administered as prescribed by the treating clinician in standard real-world practice. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Neoadjuvant duration as determined by the clinician prior to surgery, with total outcomes followed for up to 42 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care surgical resection, radiation, and subsequent adjuvant therapies corresponding to the neoadjuvant treatment. **Prohibited:** Any prior systemic therapies for melanoma before the initiation of the neoadjuvant index therapies.

6. Study Timeline & Visit Schedule

(Note: As a retrospective observational study, timepoints reflect real-world clinical milestones extracted from electronic medical records [EMR] rather than mandated visits).

Visit ID	Window	Key Activities/Procedures	:---	:---
:---	Index Date	Day 1	Initiation of neoadjuvant	

nivo+rela or nivo+ipi, baseline characteristics assessment. | | **Surgery** | Post-Neoadjuvant | Surgical resection of melanoma, assessment of pathologic complete response (pCR). | | **Adjuvant Phase** | Post-Surgery | Initiation of post-surgery adjuvant therapies, tracking of dose modifications. | | **End of Follow-Up** | Up to 42 Months | Assessment of EFS, OS, DMFS, TTD, and extraction of real-world AEs. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Participant baseline demographics, participant baseline clinical characteristics, and participant treatment history (e.g., number of doses of neoadjuvant therapies received, types of surgeries, time from neoadjuvant index treatment to surgery). **Measurement Method:** Retrospective EMR abstraction (Cardinal Health's OPEN network).

7.2 Secondary & Exploratory Endpoints * Pathologic Complete Response (pCR) * Overall Survival (OS) up to 42 months * Event-Free Survival (EFS) and Distant Metastasis-Free Survival (DMFS) * Time to Treatment Discontinuation (TTD) and Duration of Treatment (DOT) * Participant Adverse Events (AEs) and rationale for dose modifications/discontinuations

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Real-world AEs extracted retrospectively from EMR records over the follow-up period. **Serious Adverse Events (SAEs):** Events resulting in death, life-threatening status, or inpatient hospitalization captured via clinical notes. **Data Monitoring Committee (DMC):** N/A for retrospective observational claims/EMR studies.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Real-world cohort of adult stage III melanoma patients satisfying all inclusion and exclusion criteria. **Sample Size Justification:** Target Enrollment N=100\$ based on the availability of sufficient qualifying records in the designated database to capture representative treatment patterns. **Handling of Missing Data:** Handled using standard epidemiological and statistical techniques for retrospective database analyses (e.g., omitting incomplete fields from specific denominators).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and patient data privacy

regulations (HIPAA). **Monitoring Plan:** Systematic data validation, computerized logic checks, and quality control on abstracted EMR data. **Audit Trail:** Secure database environments with strict access controls and query logs for Cardinal Health's OPEN data abstraction.

11. Study Identifiers & Governance

Primary ID: CA224-1084 **Registry Number:** NCT07091695 **Posting ID:** 70087010672967855 **Sponsor/Lead Organization:** Bristol Myers Squibb
Study Title: Real-world Outcomes Among Patients With Melanoma Treated With Neoadjuvant Nivolumab+Relatlimab or Nivolumab+Ipilimumab **Official Regulatory Title:** Real-world Outcomes Among Patients With Melanoma Treated With Neoadjuvant Nivolumab+Relatlimab (Nivo+Rela) or Nivolumab+Ipilimumab (Nivo+Ipi)

12. Clinical Framework (Melanoma Specific)

Primary Condition: Stage III Resectable Melanoma (Malignant Melanoma) **Disease Area:** Melanoma **Preferred UMLS Name:** Immunotherapy for Melanoma **MeSH Code:** D008545 **HPO Code:** HP: 0002861 **SNOMED ID:** 372244006 **Diagnosis Threshold:** Clinically palpable Stage III disease **Medical Methodology:** Neoadjuvant Immunotherapy **Optimization Target:** Pathologic Complete Response (pCR) and Event-Free Survival (EFS)

13. Study Procedures & Methodology

Management Pathway: Real-world Neoadjuvant to Adjuvant Transition Pathway **Education Protocol (HCP):** N/A (Retrospective observational study) **Education Protocol (Patient):** N/A (Retrospective chart review) **Quality Audit Mechanism:** Electronic Medical Record (EMR) data validation and automated extraction logic checks

14. Observation & Follow-Up Schedule

Total Duration: Up to 42 months follow-up **Onsite Visit Cadence:** N/A (Real-world data collection) **Remote Monitoring Frequency:** N/A (Retrospective EMR review) **Checklist Review Interval:** Data snapshot at study conclusion

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |